
Willingness to Pay and Willingness to Accept: How Much Can They Differ?

Author(s): W. Michael Hanemann

Source: *The American Economic Review*, Jun., 1991, Vol. 81, No. 3 (Jun., 1991), pp. 635-647

Published by: American Economic Association

Stable URL: <https://www.jstor.org/stable/2006525>

JSTOR is a not-for-profit service that helps scholars, researchers, and students discover, use, and build upon a wide range of content in a trusted digital archive. We use information technology and tools to increase productivity and facilitate new forms of scholarship. For more information about JSTOR, please contact support@jstor.org.

Your use of the JSTOR archive indicates your acceptance of the Terms & Conditions of Use, available at <https://about.jstor.org/terms>



is collaborating with JSTOR to digitize, preserve and extend access to *The American Economic Review*

JSTOR

Willingness To Pay and Willingness To Accept: How Much Can They Differ?

By W. MICHAEL HANEMANN*

In many empirical studies, analysts seek to obtain money measures of welfare changes due not to price changes but to changes in the availability of public goods or amenities, changes in the qualities of commodities, or changes in the fixed quantities of rationed goods. The conventional welfare measures for price changes are the compensating (*C*) and equivalent (*E*) variations, which correspond to the maximum amount an individual would be willing to pay (WTP) to secure the change or the minimum amount she would be willing to accept (WTA) to forgo it. Karl-Göran Mäler (1974) was perhaps the first to show that the concepts of *C* and *E* can readily be extended from conventional price changes to such quantity changes. For price changes, Robert Willig (1976) demonstrated that *C* and *E* are likely to be fairly close in value, with the difference depending directly on the size of the income elasticity of demand for the commodity whose price changes. Subsequently, Alan Randall and John Stoll (1980) examined the duality theory associated with fixed quantities in the utility function and showed that, with appropriate modifications, Willig's formulas for bounds on *C* and *E* do, indeed, carry over to this setting.

Within the environmental-economics literature, Randall and Stoll's results have been widely interpreted as implying that WTP and WTA for changes in environmental amenities should not differ greatly unless there are unusual income effects.¹ However,

recent empirical work using various types of interview procedures has produced some evidence of large disparities between WTP and WTA measures.² This has led to something of an impasse: how can the empirical evidence of significant differences between WTP and WTA be reconciled with the theoretical analysis suggesting that such differences are unlikely? Can they be explained entirely by unusual income effects or by peculiarities of the interview process?

In this note, I reexamine Randall and Stoll's analysis and show that, while it is indeed accurate, its implications have been misunderstood. For quantity changes, there is no presumption that WTP and WTA must be close in value and, unlike price changes, the difference between WTP and WTA depends not only on an income effect but also on a substitution effect. By the latter, I mean the ease with which other privately marketed commodities can be substituted for the given public good or fixed commodity, while maintaining the individual at a constant level of utility. I show that, holding income effects constant, the smaller the substitution effect (i.e., the fewer substitutes available for the public good) the *greater* the disparity between WTP and WTA. This surely coincides with common intuition. If there are private goods that are readily substitutable for the public good, there ought to be little difference between an individual's WTP and WTA for a change in the public good. However, if the public good has almost no substitutes (e.g., Yosemite National Park, or in a different context, your own life), there is no reason why WTP and WTA could not differ vastly: in the limit, WTP could equal the individual's entire (finite)

*Department of Agricultural and Resource Economics, University of California, Berkeley, CA 94720.

¹This view is expressed by, for example, Myrick Freeman (1979 p. 3), Mark A. Thayer (1981 p. 30), Jack L. Knetsch and J. A. Sinden (1984 p. 508), Robin Gregory (1986 p. 326), Don L. Coursey et al. (1987 p. 678), and most of the participants in a recent symposium on valuing amenity resources edited by George L. Peterson et al. (1988 pp. 104, 129, 138, 152, 168, 230, 238, 259).

²See the summaries in table 3.2 of Ronald G. Cummings et al. (1986) and table 1 of Ann Fisher et al. (1988).

income, while WTA could be infinite. My argument is developed in the following two sections. Section I deals specifically with the two polar cases of perfect substitution and zero substitution between the public good and available private goods. Section II deals with Randall and Stoll's extension of Willig's formulas and shows that their bounds are, in fact, consistent with substantial divergences between WTP and WTA. Section III presents empirical application of these bounds and relates them to Mäler's concept of weak complementarity.

I. Two Polar Cases

The theoretical setup is as follows. An individual has preferences for various conventional market commodities whose consumption is denoted by the vector $\mathbf{x}$ as well as for another commodity whose consumption is denoted by q .³ This could represent the supply of a public good or amenity; it could be an index of the quality of one of the private goods; or it could be a private commodity whose consumption is fixed by a public agency.⁴ The key point is that the individual's consumption of q is fixed exogenously, while she can freely vary her consumption of the x 's. These preferences are represented by a utility function, $u(\mathbf{x}, q)$, which is continuous and nondecreasing in its arguments (I assume that the x 's and q are all "goods") and strictly quasiconcave in $\mathbf{x}$. The individual chooses her consumption by solving

$$(1) \quad \max_{\mathbf{x}} u(\mathbf{x}, q) \text{ subject to } \sum p_i x_i = y$$

taking the level of q as given. This yields a set of ordinary demand functions, $x_i = h^i(\mathbf{p}, q, y)$, $i = 1, \dots, N$, and an indirect utility function, $v(\mathbf{p}, q, y) \equiv u[h(\mathbf{p}, q, y), q]$,

³I am treating q as a scalar here, but it could be a vector without seriously affecting the analysis in this section. In the next section, however, the analysis would become significantly more complex if q were a vector and more than one element of q changed.

⁴These alternative interpretations are offered, respectively, by Mäler (1974), Hanemann (1982), and Randall and Stoll (1980).

which has the conventional properties with respect to the price and income arguments and also is nondecreasing in q .⁵ Now suppose that q rises from q^0 to $q^1 > q^0$ while prices and income remain constant at $(\mathbf{p}, y)$. Accordingly, the individual's utility changes from $u^0 \equiv v(\mathbf{p}, q^0, y)$ to $u^1 \equiv v(\mathbf{p}, q^1, y) \geq u^0$. Following Mäler, the compensating and equivalent variation measures of this change are defined, respectively, by⁶

$$(2) \quad v(\mathbf{p}, q^1, y - C) = v(\mathbf{p}, q^0, y)$$

$$(3) \quad v(\mathbf{p}, q^1, y) = v(\mathbf{p}, q^0, y + E).$$

Dual to the utility maximization in (1) is an expenditure minimization: minimize $\sum p_i x_i$ with respect to $\mathbf{x}$ subject to $u = (x, q)$, which yields a set of compensated demand functions, $x_i = g^i(\mathbf{p}, q, u)$, $i = 1, \dots, N$, and an expenditure function, $m(\mathbf{p}, q, u) \equiv \sum p_i g^i(\mathbf{p}, q, u)$, which has the conventional properties with respect to $(\mathbf{p}, u)$ and is non-increasing in q . In terms of this function, C and E are given by

$$(2') \quad C = m(\mathbf{p}, q^0, u^0) - m(\mathbf{p}, q^1, u^0)$$

$$(3') \quad E = m(\mathbf{p}, q^0, u^1) - m(\mathbf{p}, q^1, u^1).$$

It is evident from (2) and (3) that $0 < C < y$ while $E \geq 0$.⁷ The questions at issue are: i) is it true that $E/C \approx 1$? and ii) what factors affect this ratio? As a first cut at an answer, I compare two polar cases. In the first case, at least one private good—say, the first—is a perfect substitute for some

⁵These properties are established in my earlier paper (Hanemann, 1982).

⁶I have taken the liberty of defining C and E as the negative of quantities appearing in Willig (1976) and in Randall and Stoll (1980), so that $\text{sign}(C) = \text{sign}(E) = \text{sign}(u^1 - u^0)$.

⁷I assume throughout that $q^1 > q^0$ and $u^1 \geq u^0$. The analysis could be repeated for a case in which quality decreases and $u^1 < u^0$. In that case, C and E are both nonpositive and correspond, respectively, to the compensation that the individual would be willing to accept to consent to the change and the amount that she would be willing to pay to avoid the change. This would reverse the inequalities presented in what follows, but it would not affect the substance of my argument.

transformation of q . Thus, the direct utility function assumes the special form

$$(4) \quad u(\mathbf{x}, q) = \bar{u}[x_1 + \psi(q), x_2, \dots, x_N]$$

where $\psi(\cdot)$ is an increasing function and $\bar{u}(\cdot)$ is a continuous, increasing, strictly quasi-concave function of N variables. As W. M. Gorman (1976) has shown, assuming an interior solution, the resulting indirect utility function is

$$(5) \quad v(\mathbf{p}, q, y) = \bar{v}[p_1, p_2, \dots, p_N, y + p_1 \cdot \psi(q)]$$

where $\bar{v}(\cdot)$ is the indirect utility function corresponding to $\bar{u}(\cdot)$. Substitution of (5) into (2) and (3) yields the following:⁸

PROPOSITION 1: *If at least one private market good is a perfect substitute for q , then $C = E$.*

At the opposite extreme, I assume that there is a zero elasticity of substitution not just between q and x_1 but between q and *all* the x 's. Thus, the direct utility function becomes

$$(6) \quad u(\mathbf{x}, q) = \bar{u}\left[\min\left(q, \frac{x_1}{\alpha_1}\right), \dots, \min\left(q, \frac{x_N}{\alpha_N}\right)\right]$$

where $\alpha_1, \dots, \alpha_N$ are positive constants and $\bar{u}(\cdot)$ is conventional direct utility function. In this case, the indirect utility function $v(\mathbf{p}, q, y)$ has a rather complex structure and changes its form in different segments of $(\mathbf{p}, q, y)$ -space. It will be sufficient for my purposes to focus on just one of these segments. Suppose that $q \leq y/\sum p_i \alpha_i$; then, the maximization of (6), subject to the budget constraint, yields demand functions and an indirect utility function of the form $x_i = h^i(\mathbf{p}, q, y) = \alpha_i q$, and $u =$

$v(\mathbf{p}, q, y) = \bar{u}(q, \dots, q) \equiv w(q)$. In this region of $(\mathbf{p}, q, y)$ -space, the individual does not exhaust her budget, and her marginal utility of income is therefore zero. Now suppose that $q^0 \leq y/\sum p_i \alpha_i$ and $q^1 > q^0$. Since $v(\mathbf{p}, q^1, y) > w(q^0)$, it is evident from (2) that the individual would be willing to pay some positive but limited amount C to secure this change. However, for any positive quantity E , no matter how large, $v(\mathbf{p}, q^0, y + E) = v(\mathbf{p}, q^0, y) = w(q^0)$. This implies the following proposition.

PROPOSITION 2: *If there is zero substitutability between q and each of the private-market goods, it can happen that, while the individual would only be willing to pay a finite amount for an increase in q , there is no finite compensation that she would accept to forgo this increase.*

It should be emphasized that this result obtains only in a portion of $(\mathbf{p}, q, y)$ space; in other regions, even with (6), E would be finite.⁹ However, the result in Proposition 2 can also be established for other utility functions that permit some substitutability between q and the x 's as long as the indifference curves between q and each of the x 's become parallel to the x axis at some point. The implication of these two propositions is that the degree of substitutability between q and private-market goods *does* significantly affect the relation between C and E . In the next section, I show how this observation can be reconciled with the bounds on C and E derived by Randall and Stoll.

II. Randall and Stoll's Bounds

In order to extend Willig's bounds from price to commodity space, Randall and Stoll focus on a set of demand functions different from those considered above. Suppose that the individual could purchase q in a market

⁸This result carries over, of course, if *more* than one private good is a perfect substitute for q . In the most general case, $u(\mathbf{x}, q) = \bar{u}[x_1 + \psi_1(q), \dots, x_N + \psi_N(q)]$ and $C = E = \sum p_i [\psi_i(q^1) - \psi_i(q^0)]$.

⁹Indeed, if $\bar{h}^i(\alpha_1 p_1, \dots, \alpha_N p_N, y) \leq q^0$, $i = 1, \dots, N$, it can be shown that $v(\mathbf{p}, q^0, y) = v(\mathbf{p}, q^1, y) = \bar{v}(\alpha_1 p_1, \dots, \alpha_N p_N, y)$ and $C = E = 0$, where $\bar{h}^i(\cdot)$ and $\bar{v}(\cdot)$ are the ordinary demand functions and the indirect utility function associated with $\bar{u}(\cdot)$.

at some given price, π . It must be emphasized that this market is entirely hypothetical since q is actually a public good. Instead of (1), she would now solve¹⁰

$$(7) \quad \max_{\mathbf{x}, q} u(\mathbf{x}, q)$$

subject to $\sum p_i x_i + \pi q = y$.

Denote the resulting ordinary demand functions by $x_i = \hat{h}^i(\mathbf{p}, \pi, y)$, $i = 1, \dots, N$ and $q = \hat{h}^q(\mathbf{p}, \pi, y)$. The corresponding indirect utility function is $\hat{v}(\mathbf{p}, \pi, y) \equiv u[\hat{h}(\mathbf{p}, \pi, y), \hat{h}^q(\mathbf{p}, \pi, y)]$. The dual to (7) is: minimize $\sum p_i x_i + \pi q$ with respect to $\mathbf{x}$ and q subject to $u = u(\mathbf{x}, q)$. This generates a set of compensated demand functions, $x_i = \hat{g}^i(\mathbf{p}, \pi, u)$, $i = 1, \dots, N$ and $q = \hat{g}^q(\mathbf{p}, \pi, u)$, and an expenditure function, $\hat{m}(\mathbf{p}, \pi, u) \equiv \sum p_i \hat{g}^i(\mathbf{p}, \pi, u) + \pi \hat{g}^q(\mathbf{p}, \pi, u)$. These functions are hypothetical, since q is really exogenous to the individual, but they are of theoretical interest because they shed light on the relation between C and E .

For any given values of q , $\mathbf{p}$, and u , the equation

$$(8) \quad q = \hat{g}^q(\mathbf{p}, \pi, u)$$

may be solved to obtain $\pi = \hat{\pi}(\mathbf{p}, q, u)$, the inverse compensated demand (i.e., willingness-to-pay) function for q : $\hat{\pi}(\cdot)$ is the price that would induce the individual to purchase q units of the public good in order to attain a utility level of u , given that she could buy private goods at prices $\mathbf{p}$. Let $\pi^0 \equiv \hat{\pi}(\mathbf{p}, q^0, u^0)$ and $\pi^1 \equiv \hat{\pi}(\mathbf{p}, q^1, u^1)$ denote the prices that would have supported q^0 and q^1 , respectively. The two expenditure functions dual to (1) and (7) are related by

$$(9) \quad m(\mathbf{p}, q, u) \equiv \hat{m}[\mathbf{p}, \hat{\pi}(\mathbf{p}, q, u), u] \\ - \hat{\pi}(\mathbf{p}, q, u) \cdot q.$$

¹⁰It is now necessary to assume that $u(\cdot)$ is strictly quasi-concave in both $\mathbf{x}$ and q , rather than $\mathbf{x}$ alone. See footnote 22 for an example in which this is a nontrivial restriction.

This implies that¹¹

$$(10) \quad m_q(\mathbf{p}, q, u) = -\hat{\pi}(\mathbf{p}, q, u).$$

Combining (10) with (2') and (3') yields these alternative formulas for C and E , expressed in terms of the willingness-to-pay function:

$$(2'') \quad C = \int_{q^0}^{q^1} \hat{\pi}(\mathbf{p}, q, u^0) dq$$

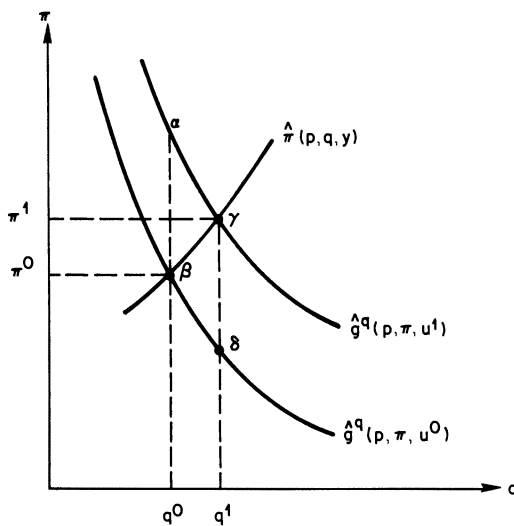
$$(3'') \quad E = \int_{q^0}^{q^1} \hat{\pi}(\mathbf{p}, q, u^1) dq.$$

It can be shown that $\text{sign}(\hat{\pi}_u) = \text{sign}(\hat{h}_y^q)$. Therefore, for given $(\mathbf{p}, q)$, the graph of $\hat{\pi}(\mathbf{p}, q, u^1)$ lies above (below) that of $\hat{\pi}(\mathbf{p}, q, u^0)$, and $E > (<) C$, when q is a normal (inferior) good. Figure 1 shows E and C for the case in which q is normal: E corresponds to the area $q^0 \alpha \gamma q^1$, while C corresponds to the area $q^0 \beta \delta q^1$.

Using techniques pioneered by Willig (1976), Randall and Stoll (1980) establish bounds on the difference between each of C and E and the area under an inverse ordinary demand function for q . From this, they derive bounds on the difference between C and E . However, the requisite inverse ordinary demand function is obtained in a rather special manner. Given any level of q , one can ask what market price π would induce the individual to purchase that amount of public good if it were available in a market, while still allowing her to purchase the quantity of the x 's that she actually did buy at market prices $\mathbf{p}$ and with income y . In conducting this thought experiment, one needs to supplement the individual's income so that she can afford q as well as the x 's. Thus, for given $(\mathbf{p}, q, y)$, one seeks the price π that satisfies

$$(11) \quad q = \hat{h}^q(\mathbf{p}, \pi, y + \pi q).$$

¹¹Using subscripts to denote derivatives, differentiate (9) and note that $q = \hat{g}^q(\mathbf{p}, \pi, u) = \hat{m}_\pi(\mathbf{p}, \pi, u)$ by Shephard's lemma. Equations similar to (9)–(12) are presented by J. P. Neary and K. W. S. Roberts (1980).

FIGURE 1. WTP AND WTA FOR A CHANGE IN q

The solution will be denoted by $\pi = \hat{\pi}(\mathbf{p}, q, y)$. This inverse function is related to the inverse compensated demand function by the identities¹²

$$(12a) \quad \hat{\pi}(\mathbf{p}, q, y) \\ \equiv \hat{\pi}[\mathbf{p}, q, v(\mathbf{p}, q, y)]$$

$$(12b) \quad \hat{\pi}(\mathbf{p}, q, u) \\ \equiv \hat{\pi}[\mathbf{p}, q, m(\mathbf{p}, q, u)].$$

Both identities play a role in the analysis. From (12a) it follows that $\pi^0 \equiv \hat{\pi}(\mathbf{p}, q^0, u^0) \equiv \hat{\pi}(\mathbf{p}, q^0, y)$ and $\pi^1 \equiv \hat{\pi}(\mathbf{p}, q^1, u^1) \equiv \hat{\pi}(\mathbf{p}, q^1, y)$. Hence, the graph of $\hat{\pi}(\mathbf{p}, q, y)$ as a function of q intersects the graph of $\hat{\pi}(\mathbf{p}, q, u^0)$ at $q = q^0$, and the graph of $\hat{\pi}(\mathbf{p}, q, u^1)$ at $q = q^1$. This is depicted in

¹²Note that $\hat{\pi}(\mathbf{p}, q, y)$ is *not* an inverse ordinary demand function in the sense of Ronald W. Anderson (1980), because it involves an income adjustment as well as a price effect.

Figure 1.¹³ From (12b) and (10), one obtains

$$(13) \quad m_q(\mathbf{p}, q, u) \\ = -\hat{\pi}[\mathbf{p}, q, m(\mathbf{p}, q, u)]$$

which is the fundamental differential equation underlying Randall and Stoll's analysis.¹⁴ Also, by differentiating (12b) and then using (13), it can be shown that the concavity of $\hat{m}(\mathbf{p}, \pi, u)$ in π , which itself follows from the quasi-concavity of $u(\mathbf{x}, q)$ in q , implies the following negativity condition on the Slutsky term associated with $\hat{\pi}(\mathbf{p}, q, y)$: $\hat{\pi}_q - \pi \hat{\pi}_y \leq 0$.¹⁵

Define the quantity

$$(14) \quad A \equiv \int_{q^0}^{q^1} \hat{\pi}(\mathbf{p}, q, y) dq$$

which corresponds to the area $q^0\beta\gamma\delta q^1$ in Figure 1. This is a sort of Marshallian consumer's surplus, which is to be compared with C and E .¹⁶ Let

$$\xi \equiv \frac{\partial \ln \hat{\pi}(\mathbf{p}, q, y)}{\partial \ln y}$$

be the income elasticity of $\hat{\pi}(\mathbf{p}, q, y)$; Randall and Stoll call this the "price flexibility of income." Assume that, over the range from $(\mathbf{p}, q^0, y)$ to $(\mathbf{p}, q^1, y)$, this elasticity is bounded from below by ξ^L and from above by ξ^U , with neither bound equal to 1. Using

¹³It is commonly supposed the $\hat{\pi}_q < 0$, so that $\pi^0 > \pi^1$ when $q^0 < q^1$ (see e.g., Richard E. Just et al., 1982 fig. 7.12), but this is not correct. It can be shown that $\pi^0 \geq \pi^1$ according to whether $\eta \leq 1/\alpha$, where η and α are defined in the text below equation (16'). Since $\sum \alpha_i \eta_i + \alpha \eta = 1$ by the Engel aggregation condition, where $\alpha_i \equiv p_i \hat{h}'_i(\mathbf{p}, \pi, y + \pi q) / (y + \pi q)$ and $\eta_i \equiv (y + \pi q) \hat{h}'_y(\mathbf{p}, \pi, y + \pi q) / \hat{h}(\mathbf{p}, \pi, y + \pi q)$, it follows that $\pi^0 \leq \pi^1$ if and only if $\sum \alpha_i \eta_i \leq 0$.

¹⁴This corresponds to Randall and Stoll's equation 7.

¹⁵Also, because $\hat{m}(\cdot)$ is linearly homogeneous in $(\mathbf{p}, \pi)$, it follows that $\hat{\pi}(\mathbf{p}, q, y)$ is linearly homogeneous in $(\mathbf{p}, y)$.

¹⁶Its relation to the conventional Marshallian consumer's surplus associated with the demand for the x 's is analyzed in Proposition 4.

the mean-value theorem, as in Willig's (1976) equation 18, and integrating (13) yields Randall and Stoll's result, namely, the following.

PROPOSITION 3: Assume $\xi^L \leq \xi \leq \xi^U$ where $\xi^L \neq 1$ and $\xi^U \neq 1$. Then,

$$(i) \quad 0 \leq \left[1 + (1 - \xi^L) \frac{A}{y} \right]^{\frac{1}{1 - \xi^L}} - 1 \leq \frac{E}{y}$$

$$(ii) \quad 0 \leq 1 - \left[1 - (1 - \xi^U) \frac{A}{y} \right]^{\frac{1}{1 - \xi^U}} \leq \frac{C}{y} \leq 1.$$

If $\xi^U < 1$, or if $\xi^U > 1$ and $1 + (1 - \xi^U) \frac{A}{y} > 0$,

$$(iii) \quad \frac{E}{y} \leq \left[1 + (1 - \xi^U) \frac{A}{y} \right]^{\frac{1}{1 - \xi^U}} - 1.$$

If $\xi^L > 1$, or if $\xi^L \leq 1$ and $1 - (1 - \xi^L) \frac{A}{y} \geq 0$,

$$(iv) \quad \frac{C}{y} \leq 1 - \left[1 - (1 - \xi^L) \frac{A}{y} \right]^{\frac{1}{1 - \xi^L}}.$$

Applying a Taylor approximation, as in Willig (1976), and assuming that the conditions in (iii) and (iv) are satisfied, one obtains

$$(15) \quad \xi^L \frac{A^2}{2y} \leq E - C \leq \xi^U \frac{A^2}{2y}.$$

This is commonly interpreted as implying that C and E are likely to be close in value, but whether or not that is correct clearly depends on the magnitudes of A/y and the bounds ξ^L and ξ^U . The magnitude of A/y depends in part on the size of the change from q^0 to q^1 ; but what can be said about the likely magnitude of the income elasticity, ξ ? Could it happen, for example, that $\xi^L = \infty$? To answer that question, dif-

ferentiate (11) implicitly:

$$(16) \quad \frac{\partial \hat{\pi}(\mathbf{p}, q, y)}{\partial y} = - \frac{\hat{h}_y^q(\mathbf{p}, \pi, y + \pi q)}{\hat{h}_\pi^q(\mathbf{p}, \pi, y + \pi q) + q \hat{h}_y^q(\mathbf{p}, \pi, y + \pi q)}.$$

By the Hicks-Slutsky decomposition, the denominator is equal to the own-price derivative of the compensated demand function for q and is nonpositive:

$$\begin{aligned} \hat{g}_\pi^q[\mathbf{p}, \pi, v(\mathbf{p}, q, y)] \\ = \hat{h}_\pi^q(\mathbf{p}, \pi, y + \pi q) + q \hat{h}_y^q(\mathbf{p}, \pi, y + \pi q) \\ \leq 0. \end{aligned}$$

Converted to elasticity form, (16) becomes

$$(16') \quad \xi = - \frac{\eta(1 - \alpha)}{\varepsilon}$$

where

$$\eta \equiv \frac{(y + \pi q) \hat{h}_y^q(\mathbf{p}, \pi, y + \pi q)}{\hat{h}^q(\mathbf{p}, \pi, y + \pi q)}$$

is the income elasticity of the *direct* ordinary demand function for q ,

$$\alpha \equiv \frac{\pi \hat{h}^q(\mathbf{p}, \pi, y + \pi q)}{y + \pi q}$$

is the budget share of q in relation to "adjusted" income, and

$$\varepsilon \equiv \frac{\pi \hat{g}_\pi^q[\mathbf{p}, \pi, v(\mathbf{p}, q, y)]}{\hat{g}^q[\mathbf{p}, \pi, v(\mathbf{p}, q, y)]}$$

is the own-price elasticity of the compensated demand function for q .

The denominator in (16') can be related to the overall elasticity of substitution be-

tween q and the private market goods $x_1, \dots, x_N$. Assume that the prices $p_1, \dots, p_N$ vary in strict proportion (i.e., $p_i = \theta \bar{p}_i$ for some fixed vector $\bar{\mathbf{p}}$ and some positive scalar θ). Let the aggregate Allen-Uzawa elasticity of substitution between q and the Hicksian composite commodity $x_0 \equiv \sum \bar{p}_i x_i$ be denoted by σ_0 . By adapting W. E. Diewert's (1974) analysis, the following formula can be established relating σ_0 to the compensated own-price elasticity for q : $\varepsilon = -\sigma_0(1 - \alpha)$.¹⁷ Hence, (16') may be written

$$(17) \quad \xi = \frac{\eta}{\sigma_0}.$$

This equation is my fundamental result. It explains the findings in the preceding section about the importance of substitution elasticities. It demonstrates that for changes in q , unlike for changes in p , the extent of the difference between C and E depends not only on income effects (i.e., η) but also on substitution effects (i.e., σ_0). If, over the relevant range, either $\eta = 0$ (no income effects) or $\sigma_0 = \infty$ (perfect substitution between q and one or more of the x 's), then $\xi^L = \xi^U = 0$ and, from Proposition 3, $C = A = E$. On the other hand, if either the demand function for q is highly income elastic or there are very few substitutes for q among the x 's so that σ_0 is close to zero, this could generate very large values of ξ and a substantial divergence between C and E .

¹⁷In deriving this result, one evaluates $\hat{m}(\mathbf{p}, \pi, u)$, $\hat{g}^q(\mathbf{p}, \pi, u)$, and $\hat{g}(\mathbf{p}, \pi, u)$ at $u = v(\mathbf{p}, q, y)$. Hence, the budget shares introduced above satisfy $\alpha \equiv \pi \hat{h}^q(\mathbf{p}, \pi, y + \pi q) / (y + \pi q) = \pi \hat{g}^q(\mathbf{p}, \pi, u) / \hat{m}(\mathbf{p}, \pi, u)$, and $\alpha_j \equiv p_j \hat{h}^j(\mathbf{p}, \pi, y + \pi q) / (y + \pi q) = p_j \hat{g}^j(\mathbf{p}, \pi, u) / \hat{m}(\mathbf{p}, \pi, u)$. In addition to the compensated own-price demand elasticity, ε , I introduce the compensated cross-price demand elasticities $\varepsilon_j \equiv \partial \ln \hat{g}^q(\mathbf{p}, \pi, u) / \partial p_j$. The homogeneity of $\hat{g}^q(\cdot)$ in $(\mathbf{p}, \pi)$ implies that $\varepsilon + \sum \varepsilon_j = 0$. Under the assumption that $p_j = \theta \bar{p}_j$, $j = 1, \dots, N$, the Allen-Uzawa elasticity of substitution between q and the composite good x_0 is given by $\sigma_0 \equiv [(\partial \hat{g}^q / \partial \theta) \cdot \theta / q] / \alpha_0$ where $\alpha_0 \equiv \sum \alpha_j = 1 - \alpha$ is the expenditure share of x_0 and θ is treated as its price. Observe that $\partial \hat{g}^q / \partial \theta = \sum (\partial \hat{g}^q / \partial p_j) \cdot \bar{p}_j = \sum (\partial \hat{g}^q / \partial p_j) \cdot (p_j / \theta) = (q / \theta) \sum \varepsilon_j = -(q / \theta) \varepsilon$. Hence, $\sigma_0(1 - \alpha) = (\theta / q)(\partial \hat{g}^q / \partial \theta) = -\varepsilon$.

III. Applications

In the first application of (17), the price flexibility of income, ξ , is assumed to be a constant. In that case, as Randall and Stoll note, the bounds in Proposition 3 hold as equalities, and (for simplicity I focus on the case where $\xi \neq 1$):

$$(18a) \quad \frac{C}{y} = 1 - \left[1 - (1 - \xi) \frac{A}{y} \right]^{\frac{1}{1 - \xi}}$$

$$(18b) \quad \frac{E}{y} = \left[1 + (1 - \xi) \frac{A}{y} \right]^{\frac{1}{1 - \xi}} - 1.$$

However, before these formulas can be used to calculate C and E , one must determine how A varies with ξ .

If the price flexibility of income is to be constant, the inverse ordinary demand function must take the form

$$(18c) \quad \hat{\pi}(\mathbf{p}, q, y) = \psi(\mathbf{p}, q) y^\xi$$

where $\psi \geq 0$. Define $G(\mathbf{p}, q) \equiv \int \psi(\mathbf{p}, q) dq$; from (14):¹⁸

$$(18d) \quad A = y^\xi [G(\mathbf{p}, q^1) - G(\mathbf{p}, q^0)].$$

Substituting (18c) into (13) and integrating yields the indirect utility function that generates (18c):

$$(18e) \quad v(\mathbf{p}, q, y)$$

$$= T \left(\left[y^{1 - \xi} + (1 - \xi) G(\mathbf{p}, q) \right]^{\frac{1}{1 - \xi}}, \mathbf{p} \right)$$

where $T(\cdot)$ is some function that is homogeneous of degree zero, increasing in its first argument, and nonincreasing in its other arguments.

¹⁸In order to satisfy the negativity and homogeneity conditions, $\psi(\cdot)$ should be increasing in q and homogeneous of degree $1 - \xi$ in $\mathbf{p}$. It follows that $G(\cdot)$, too, is homogeneous of degree $1 - \xi$ in $\mathbf{p}$.

The corresponding demand function for q , $\hat{h}^q(\mathbf{p}, \pi, y)$ can be derived from (18e) by solving $\max_q [v(\mathbf{p}, q, y - \pi q)]$. In general, a closed-form solution cannot be obtained. However, implicit differentiation of the first-order condition for this maximization yields the following expression for the income elasticity of demand:

$$\eta = \frac{\xi \psi + \xi \psi^2 q y^{\xi-1}}{-q \psi_q + \xi \psi^2 q y^{\xi-1}}.$$

It follows that having a constant ξ is generally *not* consistent with having a constant η or a constant σ_0 . An exception occurs when

$$(19a) \quad \xi \psi \equiv -q \psi_q$$

in which case $\eta \equiv 1$ and, from (17), $\sigma_0 \equiv 1/\xi$ (i.e., the price flexibility of income is merely the reciprocal of the elasticity of substitution between q and the x 's). Integrating (19a) yields $\psi(\mathbf{p}, q) = K(\mathbf{p})q^{-\xi}$ for some function $K(\mathbf{p}) \geq 0$ which is homogeneous of degree $1-\xi$, and $G(\mathbf{p}, q) = K(\mathbf{p})q^{1-\xi}/(1-\xi)$. Hence,

$$(19b) \quad A/y = K(\mathbf{p})y^{\xi-1} \left[(q^1)^{1-\xi} - (q^0)^{1-\xi} \right] / (1-\xi)$$

and the formulas for C and E become

$$(19c) \quad C/y = 1 - \left\{ 1 - K(\mathbf{p})y^{\xi-1} \left[(q^1)^{1-\xi} - (q^0)^{1-\xi} \right] \right\}^{\frac{1}{1-\xi}}$$

$$(19d) \quad E/y = \left\{ 1 + K(\mathbf{p})y^{\xi-1} \left[(q^1)^{1-\xi} - (q^0)^{1-\xi} \right] \right\}^{\frac{1}{1-\xi}} - 1.$$

This model with a unitary income elasticity, η , is the only case in which ξ , η , and σ_0 can all be constant simultaneously. It can be shown to be a generalization of the CES utility model $u(\mathbf{x}, q) =$

$[\phi(\mathbf{x})^{1-\xi} + aq^{1-\xi}]^{1/1-\xi}$ with homogeneous aggregator function, $\phi(\cdot)$.¹⁹

Equation (19b) makes explicit the dependence of A on ξ . From (19c) and (19d), it follows that the ratio E/C is increasing in both $K(\mathbf{p})$ and ξ . Table 1 tabulates this ratio for several values of K and ξ , for cases where $q^0 = 1$ and $q^1 = 3$. Observe that a low elasticity of substitution ($\sigma_0 \approx 0.07$) can generate a fivefold difference between C and E , even when A/y is very small.²⁰ A similar divergence between C and E can be obtained with a relatively moderate elasticity of substitution ($\sigma_0 = 0.99$), provided that the change matters a lot, in the sense that C/y is large ($C/y \approx 0.8$). However, C is almost identical to E when moderate or large elasticities of substitution are combined with low values of C/y .

The second application is the case in which the inverse demand function takes the form

$$(20a) \quad \hat{\pi}(\mathbf{p}, q, y) = \psi(\mathbf{p}, q)e^{\gamma(\mathbf{p})y}$$

and the price flexibility of income is $\xi = \gamma(\mathbf{p}) \cdot y$, for some $\gamma(\cdot) \geq 0$ which is homogeneous of degree -1 in $\mathbf{p}$, and some $\psi(\cdot) \geq 0$ which is homogeneous of degree 1 in $\mathbf{p}$. Substituting (20a) into (13) and integrating yields the indirect utility function that generates (20a):

$$(20b) \quad v(\mathbf{p}, q, y) = T \left(-\frac{e^{-\gamma(\mathbf{p})y}}{\gamma(\mathbf{p})} + G(\mathbf{p}, q), \mathbf{p} \right)$$

¹⁹The difference is that the CES model generates an indirect utility function of the form

$$\tilde{v}(\mathbf{p}, q, y) \equiv [y^{1-\xi} + K(\mathbf{p})q^{1-\xi}]^{1/1-\xi}$$

whereas the indirect utility function associated with (19a)–(19c) is

$$v(\mathbf{p}, q, y) = T[\tilde{v}(\mathbf{p}, q, y), \mathbf{p}].$$

²⁰This is the order of magnitude by which WTA exceeds WTP in some of the empirical studies summarized in table 3.2 of Cummings et al. (1986).

TABLE 1—SIMULATIONS OF WTP AND WTA FOR A GENERALIZED CES UTILITY MODEL

ξ	y	$K(\mathbf{p})$	σ_0	A/y	C/y	E/y	E/C
14	1	0.95	0.0714	0.073	0.05	0.259	5.175
1.01	100	1.4	0.99	1.602	0.796	4.026	5.059
0.677	100	8.1	1.481	2.414	0.991	4.975	5.003
0.677	100	0.1	1.481	0.03	0.029	0.03	1.02

where $G \equiv \int \psi(\mathbf{p}, q) dq$, and T is some function that is homogeneous of degree zero, increasing in its first argument, and nonincreasing in the other arguments. The corresponding formula for the elasticity of substitution between q and the x 's, expressed as a function of $(\mathbf{p}, q, y)$, can be shown to be

$$(20c) \quad \sigma_0 = \frac{\psi y + q\psi^2 e^{\gamma y}}{-qy\psi_q + \gamma yq\psi^2 e^{\gamma y}}.$$

From (20a) and (20b) it follows that

$$(20d) \quad A = e^{\gamma(\mathbf{p})y} [G(\mathbf{p}, q^1) - G(\mathbf{p}, q^0)]$$

$$(20e) \quad C = \gamma(\mathbf{p})^{-1} \cdot \ln[1 + \gamma(\mathbf{p})A]$$

$$(20f) \quad E = -\gamma(\mathbf{p})^{-1} \cdot \ln[1 - \gamma(\mathbf{p})A].$$

(note that, when $\gamma A > 1$, $E = \infty$). The ratio E/C is clearly increasing in $\gamma(\mathbf{p})$ and A .

In order to proceed further, it is necessary to take a closer look at A . For this purpose, I focus on the special case of (20) in which $\gamma(\mathbf{p}) = \gamma/p_N$, $\psi(\mathbf{p}, q) = \delta e^{\alpha + \delta q} p_1^{1-\beta} p_N^\beta / (\beta - 1)$, and

$$(21a) \quad v(\mathbf{p}, q, y) =$$

$$T \left(-\frac{p_N}{\gamma} e^{-\gamma y/p_N} + \frac{e^{\alpha + \delta q}}{\beta - 1} p_1^{1-\beta} p_N^\beta, p_2, p_3, \dots, p_N \right)$$

with γ and δ as positive constants and with $\beta > 1$. This implies the following log-log de-

mand function for good 1:²¹

$$(21b) \quad \ln x_1 = \ln h^1(\mathbf{p}, q, y) \\ = \alpha - \beta \ln(p_1/p_N) \\ + (\gamma y/p_N) + \delta q.$$

Since $\lim_{p_1 \rightarrow \infty} \partial v(\mathbf{p}, q, y) / \partial q = 0$, good 1 is weakly complementary with q in the sense of Mäler, and C and E can be expressed in terms of the area between the compensated demand curves $g^1(\mathbf{p}, q^0, u)$ and $g^1(\mathbf{p}, q^1, u)$. Furthermore, in this model the quantity A corresponds to the change in the Marshallian consumer's surplus associated with good 1.

$$(21c) \quad A \equiv \int_{q^0}^{q^1} \hat{\pi}(\mathbf{p}, q, y) dq \\ = \left(\frac{p_1}{\beta - 1} \right) [h^1(\mathbf{p}, q^1, y) - h^1(\mathbf{p}, q^0, y)] \\ = \int_{p_1}^{\infty} [h^1(\mathbf{p}, q^1, y) - h^1(\mathbf{p}, q^0, y)] dp_1.$$

Hence, the formulas for C and E become

$$(21d) \quad \frac{C}{p_N} = \frac{1}{\gamma} \ln \left\{ 1 + \left(\frac{\gamma}{\beta - 1} \right) \left(\frac{p_1}{p_N} \right) \right. \\ \left. \times [h^1(\mathbf{p}, q^1, y) - h^1(\mathbf{p}, q^0, y)] \right\}$$

$$(21e) \quad \frac{E}{p_N} = \frac{1}{\gamma} \ln \left\{ 1 - \left(\frac{\gamma}{\beta - 1} \right) \left(\frac{p_1}{p_N} \right) \right. \\ \left. \times [h^1(\mathbf{p}, q^1, y) - h^1(\mathbf{p}, q^0, y)] \right\}.$$

²¹ It follows that, in this model, the income elasticity of demand for good 1 is equal to γy , the price flexibility of income.

TABLE 2—SIMULATIONS OF WTP AND WTA FOR A LOG-LOG UTILITY MODEL

ξ	α	δ	γ	A/y	C/y	E/y	E/C
14	-13.58	0.2	0.14	0.0695	0.0486	0.259	5.334
1.01	1.42	0.3	0.0101	0.959	0.6704	3.414	5.092
0.677	2.151	0.3	0.00677	1.427	0.9987	5.004	5.011
0.677	2.28	0.13	0.00677	0.039	0.0385	0.0396	1.027

These are tabulated in Table 2 for several values of α , γ , and δ for the case in which $p_N = 1$, $p_1 = 1.5$, $\beta = 1.125$, $y = 100$, $q^0 = 1$, and $q^1 = 3$.²² The simulations confirm that $\xi = \gamma y$ and A are the key determinants of the ratio E/C . When either ξ or A is large, then $E \gg C$; when ξ and A are both small, then $E \approx C$.

This example is a striking illustration of the power of Proposition 3. From mere inspection of the ordinary demand function for x_1 in (21b) it is hardly obvious that the term $\xi = \gamma y$ should be a key determinant of the relationship between WTP and WTA for a change in q .²³ The example also raises another issue: the possibility that the quantity A , which forms the basis for Proposition 3, may be related to the conventional Marshallian consumer's surplus associated with a private market commodity, x_1 . Under what circumstances does this carry over to other utility models? Could it, in fact, apply to the

first example based on a generalization of the CES utility model?

By way of answer to the first question, the following lemma establishes that weak complementarity is but one of two conditions that must be satisfied if A is to be equated with a change in the Marshallian consumer's surplus:

LEMMA 1: Suppose there is a private market good, say x_1 , with the properties that (a) it is nonessential and weakly complementary with q , and (b)

$$\frac{\partial \hat{\pi}(\mathbf{p}, q, y)}{\partial p_1} = - \frac{\partial h^1(\mathbf{p}, q, y)}{\partial q}.$$

Then,

$$(22) \quad A = \int_{p_1}^{\infty} [h^1(\mathbf{p}, q^1, y) - h^1(\mathbf{p}, q^0, y)] dp_1.$$

PROOF:

Weak complementarity implies $\lim_{p_1 \rightarrow \infty} m_q(\mathbf{p}, q, u) = 0$. Nonessentialness implies $\lim_{p_1 \rightarrow \infty} m(\mathbf{p}, q, u) < \infty$. Hence, by (13), condition (a) implies $\lim_{p_1 \rightarrow \infty} \hat{\pi}(\mathbf{p}, q, y) = 0$. Accordingly, one can express $\hat{\pi}(\cdot)$ as

$$\begin{aligned} \hat{\pi}(\mathbf{p}, q, y) &= - \int_{p_1}^{\infty} \frac{\partial \pi(\mathbf{p}, q, y)}{\partial p_1} dp_1 \\ &= \int_{p_1}^{\infty} \frac{\partial h^1(\mathbf{p}, q, y)}{\partial q} dp_1 \end{aligned}$$

where the second equality follows from condition (b). Invoking the definition of A in (14) and changing the order of integration yields (22).

²²The parameter values in these simulations are chosen to satisfy the inequalities $(\beta - 1)p_N \leq \gamma p_1 h^1(\mathbf{p}, q, y) \leq \beta p_N$, which ensure that the direct utility function implied by (21) is quasi-concave in $\mathbf{x}$ and q . If $u(\mathbf{x}, q)$ were not quasi-concave in q , the formulas in (21d) and (21e), would still be valid, but the Randall-Stoll bounds in Proposition 3 would not apply. That happens with another special case of (20) in which $\gamma(\mathbf{p}) = \gamma/p_N$ but $\psi(\mathbf{p}, q) = \delta p_N e^{\alpha - \beta(p_1/p_N) + \delta q/\beta}$. This generates an ordinary demand function for good 1 that is identical to (21b), except that $\ln(p_1/p_N)$ is replaced by (p_1/p_N) . Also, (21c)–(21e) hold for this model, except that $\beta - 1$ is replaced by β . However, the implicit utility function $u(\mathbf{x}, q)$ can be shown to be quasi-convex in q .

²³Also, it is hardly obvious that from the demand function in (21b) one can recover σ_0 , the elasticity of substitution between q and the x 's (all the x 's, not just x_1). This is obtained by substituting $\gamma(\mathbf{p}) \equiv \gamma/p_N$ and $\psi(\mathbf{p}, q) \equiv \delta e^{\alpha + \delta q} p_1^{1-\beta} p_N^\beta / (\beta - 1)$ into (20c). The corresponding formula for η can be obtained from $\eta = \sigma_0 \xi = \sigma_0 \gamma y / p_N$.

Application of Lemma 1 yields the answer to the second question:

PROPOSITION 4: *Partition the price vector as $\mathbf{p} = (p_1, \mathbf{p}_{(1)})$. Equation (22) holds if and only if $v(\mathbf{p}, q, y)$ can be expressed in the form*

$$(23) \quad v(\mathbf{p}, q, y) = T[G(\mathbf{p}, q), \mathbf{p}_{(1)}y]$$

where $\lim_{p \rightarrow \infty} G_p(\mathbf{p}, q) = \lim_{p_1 \rightarrow \infty} G_{p_1}(\mathbf{p}, q) = 0$.

PROOF:

Observe that the conditions on the derivatives of $G(\cdot)$ ensure that x_1 is nonessential and weakly complementary with q . The main task is to show that the functional structure in (23) is necessary and sufficient to satisfy condition (b) of Lemma 1. First use (10) and (12a), and then twice differentiate the implicit function $v(\mathbf{p}, q, y) - u = 0$ to obtain

$$\begin{aligned} \frac{\partial \hat{\pi}(\mathbf{p}, q, y)}{\partial p_1} &= -m_{qp_1}[\mathbf{p}, q, v(\mathbf{p}, q, y)] \\ &\quad - m_{qu}[\mathbf{p}, q, v(\mathbf{p}, q, y)] \cdot v_{p_1}(\mathbf{p}, q, y) \\ &= -(v_{qp_1} - v_y v_{qp_1}) / v_y^2. \end{aligned}$$

Comparing this with $\partial h^1(\mathbf{p}, q, y) / \partial q = (v_{p_1} v_{qy} - v_y v_{qp_1})^2 / v_y^2$, it can be seen that condition (b) will be satisfied if and only if $v_{qp_1} = v_y v_{qy}$. However, this is equivalent to requiring that (v_q / v_{p_1}) be independent of y , which in turn is equivalent to (23).

The log-log utility model in (21a) clearly satisfies the conditions of Proposition 4. The generalized CES utility model in (18) could also meet the conditions of the proposition, provided that p_1 appears only in the first argument of $T(\cdot)$.

Equation (23) expresses a restriction on the marginal rate of substitution between q and the price of a weakly complementary private-market good, p_1 (i.e., that it be independent of income). This condition was first introduced by Willig (1978) in his paper

on hedonic price adjustments for valuing marginal changes in q : his theorem 1 characterized the circumstances under which the marginal value of q equals the derivative with respect to q of the Marshallian consumer's surplus for x_1 , averaged over the number of units of the good consumed. Proposition 4 expresses a similar result using a different and more compact proof. Combining Propositions 3 and 4 provides a way to value *nonmarginal* changes in q by employing the change in Marshallian consumer's surplus to compute A and then using A to bound WTP or WTA. These two propositions, in effect, establish a new link between Willig's two seminal papers.²⁴

IV. Conclusion

A recent assessment of the state of the art of public-good valuation concludes "Received theory establishes that... WTP... should approximately equal... WTA.... In contrast with theoretical axioms which predict small differences between WTP and WTA, results from contingent valuation method applications wherein such measures are derived almost always demonstrate large differences between average WTP and WTA. To date, researchers have been unable to explain in any definitive way the persistently observed differences between WTP and WTA measures" (Cummings et al., 1986 p. 41).²⁵ This paper

²⁴I am very grateful to a referee for pointing out the connection with Willig's theorem 1. In my notation, Willig's theorem states that (23) is equivalent to the equality $v_q(\mathbf{p}, q, y) / v_{p_1}(\mathbf{p}, q, y) = \hat{\pi}(\mathbf{p}, q, y) / h^1(\mathbf{p}, q, y)$.

²⁵Some of the debates on divergences between WTP and WTA have focused on the concept of loss-aversion, introduced in the economics literature by Daniel Kahneman and Amos Tversky (1979). This is a different phenomenon from that involved in the Randall-Stoll bounds: it concerns the disparity between the WTP to obtain a change from q^0 to $q^0 + \Delta$ (for some $\Delta > 0$) and the WTP to avoid a change from q^0 to $q^0 - \Delta$, which is not the same as the disparity between WTP and WTA for the same change from q^0 to $q^1 = q^0 + \Delta$. However, the loss/gain disparity can be analyzed using the tools developed in this paper. In a separate paper, I have identified the conditions under which it will exceed the disparity between WTP and WTA studied here.

offers an explanation by showing that the theoretical presumption of approximate equality between WTP and WTA is misconceived. This is because, for public goods, the relation between the two welfare measures depends on a substitution effect as well as an income effect. Given that the substitution elasticity appears in the denominator of (17) and that the Engel aggregation condition places some limit on the plausible magnitude of the income elasticity in the numerator, this suggests that the substitution effects could exert a far greater leverage on the relation between WTP and WTA than the income effects. Thus, large empirical divergences between WTP and WTA may be indicative not of some failure in the survey methodology but of a general perception on the part of the individuals surveyed that the private-market goods available in their choice set are, collectively, a rather imperfect substitute for the public good under consideration.

REFERENCES

- Anderson, Ronald W., "Some Theory of Inverse Demand for Applied Demand Analysis," *European Economic Review*, November 1980, 14, 281-90.
- Coursey, Don L., Hovis, John J. and Schulze, William D., "The Disparity Between Willingness to Accept and Willingness to Pay Measures of Value," *Quarterly Journal of Economics*, August 1987, 102, 679-90.
- Cummings, Ronald G., Brookshire, David S. and Schulze, William D., *Valuing Public Goods: An Assessment of the Contingent Valuation Method*, Totowa, NJ: Rowman and Allanheld, 1986.
- Diewert, W. E., "A Note on Aggregation and Elasticities of Substitution," *Canadian Journal of Economics*, February 1974, 7, 12-20.
- Fisher, Ann, McClelland, Gary H. and Schulze, William D., "Measures of Willingness to Pay versus Willingness to Accept: Evidence, Explanations and Potential Reconciliation," in George L. Peterson, B. L. Driver, and Robin Gregory, eds., *Amenity Resource Valuation: Integrating Economics with Other Disciplines*, State College, PA: Venture, 1988.
- Freeman, A. Myrick, *The Benefits of Environmental Improvement: Theory and Practice*, Baltimore: Johns Hopkins University Press, 1979.
- Gorman, W. M., "Tricks With Utility Functions," in M. Artis and R. Nobay, eds., *Essays in Economic Analysis*, New York: Cambridge University Press, 1976, pp. 211-43.
- Gregory, Robin, "Interpreting Measures of Economic Loss: Evidence from Contingent Valuation and Experimental Studies," *Journal of Environmental Economics and Management*, December 1986, 13, 325-37.
- Hanemann, W. Michael, "Quality and Demand Analysis," in Gordon C. Rausser, ed., *New Directions in Econometric Modeling and Forecasting in U. S. Agriculture*, Amsterdam: North Holland, 1982, pp. 55-98.
- Just, Richard E., Hueth, Darrell L. and Schmitz, Andrew, *Applied Welfare Economics and Public Policy*, Englewood Cliffs, NJ: Prentice-Hall, 1982.
- Kahneman, Daniel and Tversky, Amos, "Prospect Theory: An Analysis of Decisions Under Risk," *Econometrica*, March 1979, 47, 263-91.
- Knetsch, Jack L. and Sinden, J. A., "Willingness to Pay and Compensation Demanded: Experimental Disparity in Measures of Value," *Quarterly Journal of Economics*, August 1984, 99, 507-21.
- Mäler, Karl-Göran, *Environmental Economics: A Theoretical Inquiry*, Baltimore: Johns Hopkins University Press, 1974.
- Neary, J. P. and Roberts, K. W. S., "Theory of Household Behavior Under Rationing," *European Economic Review*, January 1980, 13, 25-42.
- Peterson, George L., Driver, B. L. and Gregory, Robin, eds., *Amenity Resource Valuation: Integrating Economics with Other Disciplines*, State College, PA: Venture, 1988.
- Randall, Alan and Stoll, John R., "Consumer's Surplus in Commodity Space," *American Economic Review*, June 1980, 71, 449-57.

Thayer, Mark A., "Contingent Valuation Techniques for Assessing Environmental Impacts: Further Evidence," *Journal of Environmental Economics and Management*, March 1981, 8, 27-44.

Willig, Robert, "Consumer's Surplus Without

Apology," *American Economic Review*, September 1976, 66, 589-97.

_____, "Incremental Consumer's Surplus and Hedonic Price Adjustment," *Journal of Economic Theory*, April 1978, 17, 227-53.